RMO– 2015(1)

Max Mark – 102 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions. All Questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. In a cyclic quadrilateral ABCD, let the diagonals AC and BD intersect at X. Let the circumcircles of triangles AXD and BXC intersect again at Y. If X is the incentre of triangle ABY, show that $\angle CAD = 90^0$.
2. Let $P_1(x) = x^2 + a_1x + b_1$ and $P_2(x) = x^2 + a_2x + b_2$ be two quadratic polynomials with integer coefficients. Suppose $a_1 \neq a_2$ and there exist integers $m \neq n$ such that $P_1(m) = P_2(n)$ and $P_2(m) = P_1(n)$. Prove that $a_1 - a_2$ is even.
3. Find all fractions which can be written simultaneously in the forms $\frac{7k-5}{5k-3}$ and $\frac{6l-1}{4l-3}$, for some integers k, l.
4. Suppose 28 objects are placed along a circle at equal distances. In how many ways can 3 objects be chosen from among them so that no two of the three chosen objects are adjacent nor diametrically opposite?
5. Let ABC be a right triangle with $\angle B = 90^0$. Let E and F be respectively the mid points of AB and AC. Suppose the incentre I of triangle ABC lies on the circumcircle of triangle AEF. Find the ratio BC/AB.
6. Find all real number a such that $3 < a < 4$ and $a(a - 3\{a\})$ is an integer. (here {a} denotes the fractional part of a. For example {1.5}=0.5, {-3.4}=0.6.)

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